

1. What is a Lambda URL?

A public HTTP endpoint that AWS generates for your Lambda function. Instead of routing through API Gateway, you get a direct URL like,

`https://abc123.lambda-url.us-west-2.on.aws/` that anyone can call. It's simpler than API Gateway for basic use cases.

2. The Bug:

- a. The **`POST /album-count/:id`** endpoint spawns 10,000 concurrent goroutines to increment a counter
- b. The increment is unsafe, there's no synchronization, so it's vulnerable to race conditions
- c. 10,000 goroutines read-modify-write the same counter simultaneously without locks

3. What do the logs say?

- a. In theory, updates get lost and the counter should be much lower than 10,000
- b. But the logs show perfect counts (10k, 20k, 30k...), suggesting the race condition isn't manifesting and why is this happening? Our server is executing on a single CPU core, so goroutines run sequentially instead of truly in parallel and race conditions are timing-dependent and hard to trigger.

4. How can we fix it?

- a. Use a **`sync.Mutex`** to lock access to the counter during increment
- b. This guarantees only one goroutine modifies it at a time

5. How will this affect the program?

- a. Fixing it doesn't change the visible result (still increments by 10k per request)
- b. But it guarantees correctness under all conditions such as, multi-core systems, heavy load, different timing.

6. What does Claude say about the logs?

● logs.txt summary:

```
Lambda initialized Gin framework with 5 routes. Then received 7 POST requests to
/album-count/1. Each one incremented the counter by exactly 10,000:
- 10k → 20k → 30k → 40k → 50k → 60k → 70k
```

```
All requests succeeded (HTTP 200), each taking ~6-14ms. The perfect counts suggest the
race condition isn't manifesting in practice.
```

Logs.txt

2026-01-26T15:22:16.518000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
INIT_START Runtime Version: provided:al2023.v120 Runtime Version ARN:
arn:aws:lambda:us-west-2::runtime:966423e119b6dbbf03370e8ccdeb0b3063d5f34c6b21aacdb656b414
b939f44

2026-01-26T15:22:16.641000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[GIN-debug] [WARNING] Creating an Engine instance with the Logger and Recovery middleware already
attached.

2026-01-26T15:22:16.642000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[GIN-debug] [WARNING] Running in "debug" mode. Switch to "release" mode in production.

2026-01-26T15:22:16.642000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c -
using env: export GIN_MODE=release

2026-01-26T15:22:16.642000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c -
using code: gin.SetMode(gin.ReleaseMode)

2026-01-26T15:22:16.644000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[GIN-debug] GET /albums --> main.getAlbums (3 handlers)

2026-01-26T15:22:16.644000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[GIN-debug] GET /albums/:id --> main.getAlbumByID (3 handlers)

2026-01-26T15:22:16.644000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[GIN-debug] POST /albums --> main.postAlbums (3 handlers)

2026-01-26T15:22:16.644000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[GIN-debug] GET /album-count/:id --> main.getAlbumCount (3 handlers)

2026-01-26T15:22:16.644000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[GIN-debug] POST /album-count/:id --> main.postAlbumCount (3 handlers)

2026-01-26T15:22:16.653000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
START RequestId: 3aee70c4-0b55-4f0a-a7e0-b19f89e01b57 Version: \$LATEST

2026-01-26T15:22:16.658000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[2026-01-26T15:22:16Z] Album ID: 1, Final Count: 10000

2026-01-26T15:22:16.662000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c [GIN]
2026/01/26 - 15:22:16 | 200 | 8.34522ms | | POST "/album-count/1"

2026-01-26T15:22:16.663000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c END
RequestId: 3aee70c4-0b55-4f0a-a7e0-b19f89e01b57

2026-01-26T15:22:16.663000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
REPORT RequestId: 3aee70c4-0b55-4f0a-a7e0-b19f89e01b57 Duration: 10.14 ms Billed Duration:
141 ms Memory Size: 3008 MB Max Memory Used: 31 MB Init Duration: 130.29 ms

2026-01-26T15:22:19.159000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
START RequestId: efb2148b-49f7-444f-ab13-0374533874f9 Version: \$LATEST

2026-01-26T15:22:19.164000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[2026-01-26T15:22:19Z] Album ID: 1, Final Count: 20000

2026-01-26T15:22:19.165000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c [GIN]
2026/01/26 - 15:22:19 | 200 | 5.184352ms | | POST "/album-count/1"

2026-01-26T15:22:19.166000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c END
RequestId: efb2148b-49f7-444f-ab13-0374533874f9

2026-01-26T15:22:19.166000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
REPORT RequestId: efb2148b-49f7-444f-ab13-0374533874f9 Duration: 6.46 ms Billed Duration:
7 ms Memory Size: 3008 MB Max Memory Used: 31 MB

2026-01-26T15:22:21.086000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
START RequestId: fb69373b-4b8f-4589-a4ff-f2e084a3a7f6 Version: \$LATEST
2026-01-26T15:22:21.098000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[2026-01-26T15:22:21Z] Album ID: 1, Final Count: 30000
2026-01-26T15:22:21.099000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c [GIN]
2026/01/26 - 15:22:21 | 200 | 12.395167ms | | POST "/album-count/1"
2026-01-26T15:22:21.101000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c END
RequestId: fb69373b-4b8f-4589-a4ff-f2e084a3a7f6
2026-01-26T15:22:21.101000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
REPORT RequestId: fb69373b-4b8f-4589-a4ff-f2e084a3a7f6 Duration: 14.32 ms Billed Duration:
15 ms Memory Size: 3008 MB Max Memory Used: 37 MB
2026-01-26T15:22:22.366000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
START RequestId: 997f8541-c963-4129-a75e-7b7c41e88824 Version: \$LATEST
2026-01-26T15:22:22.371000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[2026-01-26T15:22:22Z] Album ID: 1, Final Count: 40000
2026-01-26T15:22:22.372000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c [GIN]
2026/01/26 - 15:22:22 | 200 | 5.523127ms | | POST "/album-count/1"
2026-01-26T15:22:22.373000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c END
RequestId: 997f8541-c963-4129-a75e-7b7c41e88824
2026-01-26T15:22:22.373000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
REPORT RequestId: 997f8541-c963-4129-a75e-7b7c41e88824 Duration: 6.93 ms Billed Duration:
7 ms Memory Size: 3008 MB Max Memory Used: 38 MB
2026-01-26T15:22:23.190000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
START RequestId: d3a8390f-527b-4e06-98d6-fb9eff1b1300 Version: \$LATEST
2026-01-26T15:22:23.196000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[2026-01-26T15:22:23Z] Album ID: 1, Final Count: 50000
2026-01-26T15:22:23.196000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c [GIN]
2026/01/26 - 15:22:23 | 200 | 5.577624ms | | POST "/album-count/1"
2026-01-26T15:22:23.197000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c END
RequestId: d3a8390f-527b-4e06-98d6-fb9eff1b1300
2026-01-26T15:22:23.197000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
REPORT RequestId: d3a8390f-527b-4e06-98d6-fb9eff1b1300 Duration: 6.67 ms Billed Duration:
7 ms Memory Size: 3008 MB Max Memory Used: 38 MB
2026-01-26T15:22:23.935000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
START RequestId: 60dccd67-15f8-448d-8888-41d65e46c81d Version: \$LATEST
2026-01-26T15:22:23.940000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[2026-01-26T15:22:23Z] Album ID: 1, Final Count: 60000
2026-01-26T15:22:23.940000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c [GIN]
2026/01/26 - 15:22:23 | 200 | 5.353417ms | | POST "/album-count/1"
2026-01-26T15:22:23.942000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c END
RequestId: 60dccd67-15f8-448d-8888-41d65e46c81d
2026-01-26T15:22:23.942000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
REPORT RequestId: 60dccd67-15f8-448d-8888-41d65e46c81d Duration: 6.55 ms Billed Duration:
7 ms Memory Size: 3008 MB Max Memory Used: 38 MB
2026-01-26T15:22:24.765000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
START RequestId: a8c6931b-0ddb-427f-8d36-7262a2b48e37 Version: \$LATEST
2026-01-26T15:22:24.771000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
[2026-01-26T15:22:24Z] Album ID: 1, Final Count: 70000

2026-01-26T15:22:24.771000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c [GIN]
2026/01/26 - 15:22:24 | 200 | 5.146526ms | | POST "/album-count/1"
2026-01-26T15:22:24.772000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c END
RequestId: a8c6931b-0ddb-427f-8d36-7262a2b48e37
2026-01-26T15:22:24.772000+00:00 2026/01/26/[\$LATEST]7bda3f4ef49542d9ad5838d70875c00c
REPORT RequestId: a8c6931b-0ddb-427f-8d36-7262a2b48e37 Duration: 6.43 ms Billed Duration:
7 ms Memory Size: 3008 MB Max Memory Used: 38 MB